

PE-EC801A — Antennas and Propagation

Semester VIII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Physical Concept of Radiation	2
1.1	Mechanism of Radiation	2
1.2	Radiation from Single Wire, Two Wires, and Open-End Transmission Line	2
2	Radiation Pattern and Field Regions	3
2.1	Lobe Structure of Radiation Pattern	3
2.2	Half-Power Beamwidth (HPBW) and First-Null Beamwidth (FNBW)	3
2.3	Antenna Field Regions	3
3	Reciprocity, Directivity, and Gain	4
3.1	Reciprocity Theorem	4
3.2	Directivity and Gain	5
4	Effective Aperture and Polarization	5
4.1	Effective Aperture (A_e)	5
4.2	Polarization	6
5	Input Impedance, Efficiency, and Friis Transmission Equation	6
5.1	Input Impedance	6
5.2	Antenna Efficiency	6
5.3	Friis Transmission Equation	7
6	Radiation Integrals and Potential Functions	7
6.1	Relationship between Potentials and Fields	8
6.2	Inhomogeneous Helmholtz Wave Equation	8

1. Physical Concept of Radiation

Definition of an Antenna

An **antenna** is a metallic device (transducer) that transitions guided electromagnetic waves on a transmission line into free-space electromagnetic waves in the transmitting mode, or vice versa in the receiving mode. It serves as an impedance-matching interface between transmission lines and free space.

1.1 Mechanism of Radiation

Radiation occurs when electromagnetic energy is detached from a source and propagates into free space. The basic conditions for radiation are:

- **Stationary Charge:** A stationary charge produces a static electric field, but does not radiate.
- **Uniformly Moving Charge:** A charge moving at a constant speed along a straight transmission line does not radiate (unless the boundary conditions change, e.g., the wire terminates or bends).
- **Accelerated / Decelerated Charge:** Radiation is created by a time-varying current or by an accelerated or decelerated charge.

The acceleration of charge a_z is related to time-varying current by:

$$\frac{dI_z}{dt} = q \frac{dv_z}{dt} = qa_z$$

where q is the charge and v_z is the velocity. If a current changes over time, charges accelerate, which creates radiation.

1.2 Radiation from Single Wire, Two Wires, and Open-End Transmission Line

- **Single Wire:** If a charge is accelerated along a curved wire, it radiates.
- **Two Wires (Transmission Line):** In a two-wire parallel transmission line, the spacing between the conductors is much smaller than the wavelength ($d \ll \lambda$). The currents flow in opposite directions, so the radiated fields from the two wires cancel out in the far field. Consequently, the line acts purely as a guide.
- **Bended/Flared Line:** If the two wires are flared outward (bent), the currents are no longer anti-parallel. Their fields do not cancel, and energy is released as radiated waves.

2. Radiation Pattern and Field Regions

Definition of Radiation Pattern

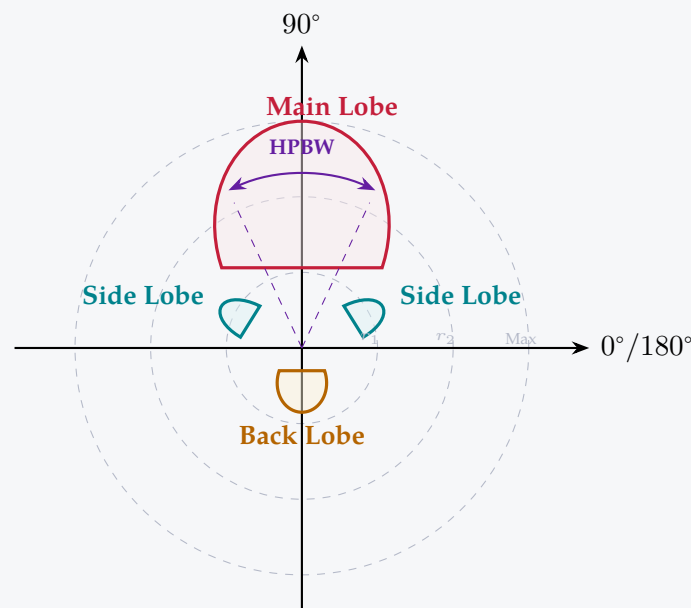
A **radiation pattern** (or antenna pattern) is a mathematical function or graphical representation of the radiation properties of the antenna as a function of space coordinates (typically spherical coordinates θ, ϕ). These properties include power flux density, field intensity, phase, and polarization.

2.1 Lobe Structure of Radiation Pattern

An antenna radiation pattern typically consists of various lobes:

- **Major Lobe (Main Lobe):** The radiation lobe containing the direction of maximum radiation.
- **Minor Lobe:** Any lobe other than the major lobe. These represent unwanted radiation.
- **Side Lobe:** A minor lobe in any direction other than the intended lobe.
- **Back Lobe:** A minor lobe whose axis is approximately 180° from the main beam.

Antenna Radiation Pattern — Lobe Structure



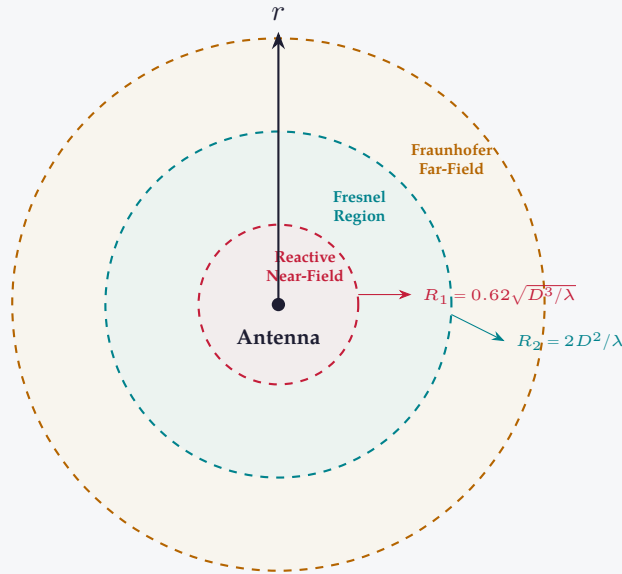
2.2 Half-Power Beamwidth (HPBW) and First-Null Beamwidth (FNBW)

- **Half-Power Beamwidth (HPBW):** The angular width of the main beam between two directions where the radiation intensity is half (3 dB below) the maximum value.
- **First-Null Beamwidth (FNBW):** The angular width between the first two nulls (where the radiation intensity drops to zero) on either side of the main beam.

2.3 Antenna Field Regions

The space surrounding an antenna is divided into three regions depending on the distance r from the antenna. Let D be the maximum dimension of the antenna, and λ be the wavelength.

Antenna Field Regions



1. **Reactive Near-Field Region:** The region immediately surrounding the antenna where the reactive fields (non-radiating, storing energy like a capacitor or inductor) are dominant. The boundary is:

$$r < R_1 = 0.62 \sqrt{\frac{D^3}{\lambda}}$$

2. **Radiating Near-Field (Fresnel) Region:** The region between the reactive near-field and far-field where radiation fields dominate, but the angular field distribution depends on the distance from the antenna (because phase variations across the aperture are significant). The boundary is:

$$0.62 \sqrt{\frac{D^3}{\lambda}} \leq r < R_2 = \frac{2D^2}{\lambda}$$

3. **Far-Field (Fraunhofer) Region:** The region where the angular distribution of the electromagnetic fields is independent of the distance. The wave behaves as a local transverse electromagnetic (TEM) plane wave, where the electric and magnetic fields are orthogonal to the direction of propagation. The boundary is:

$$r \geq \frac{2D^2}{\lambda} \quad (\text{provided } D \gg \lambda \text{ and } r \gg \lambda)$$

3. Reciprocity, Directivity, and Gain

3.1 Reciprocity Theorem

The reciprocity theorem states that the electrical characteristics (such as radiation pattern, directivity, input impedance, and effective length) of an antenna remain identical whether it is operated as a transmitter or a receiver.

Mathematical Formulation: Let two antennas A and B exist in a linear, isotropic, and homogenous medium. If a voltage source V_1 applied at the terminals of antenna A produces

a current I_2 at the short-circuited terminals of antenna B , and a voltage source V_2 applied at the terminals of antenna B produces a current I_1 at the short-circuited terminals of antenna A , then:

$$\text{If } V_1 = V_2 \implies I_1 = I_2 \quad (\text{or } Z_{12} = Z_{21})$$

3.2 Directivity and Gain

- **Radiation Intensity (U):** The power radiated per unit solid angle (expressed in Watts/steradian).

$$U(\theta, \phi) = r^2 W_{rad}(\theta, \phi)$$

where W_{rad} is the radial component of the Poynting vector.

- **Directivity (D):** The ratio of the radiation intensity in a given direction to the average radiation intensity (which is equal to the intensity of an isotropic radiator).

$$D = \frac{U(\theta, \phi)}{U_0} = \frac{4\pi U(\theta, \phi)}{P_{rad}}$$

where $P_{rad} = \int_0^{2\pi} \int_0^\pi U(\theta, \phi) \sin \theta d\theta d\phi$ is the total radiated power.

- **Gain (G):** The ratio of the radiation intensity in a given direction to the radiation intensity that would be obtained if the total power accepted by the antenna (P_{in}) was radiated isotropically.

$$G = \frac{4\pi U(\theta, \phi)}{P_{in}}$$

- **Relationship between Gain and Directivity:**

$$G = \eta_{cd} D$$

where η_{cd} is the antenna radiation efficiency ($0 \leq \eta_{cd} \leq 1$), representing conduction and dielectric losses.

4. Effective Aperture and Polarization

4.1 Effective Aperture (A_e)

The **effective aperture** (or effective area) of a receiving antenna is a measure of its ability to extract power from a passing electromagnetic wave. It is defined as the ratio of the delivered power P_L at the load terminals (matched condition) to the power density W_{inc} of the incident wave.

Effective Aperture and Directivity Relation

The delivered power is $P_L = A_e W_{inc}$. The relationship between the maximum effective aperture A_{em} and the maximum directivity D_0 for any antenna is:

$$A_{em} = \frac{\lambda^2}{4\pi} D_0$$

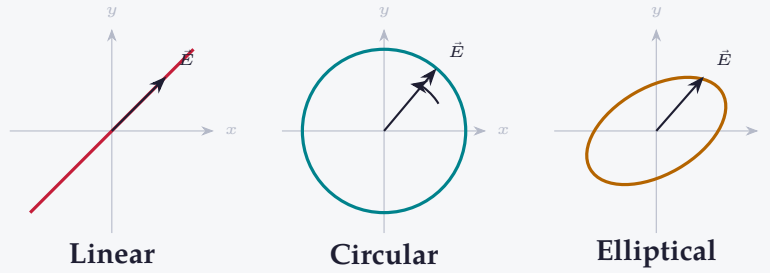
4.2 Polarization

The **polarization** of an antenna is defined as the polarization of the electromagnetic wave radiated by the antenna in a given direction. It describes the orientation of the electric field vector $\vec{E}$ as a function of time.

There are three main types of polarization:

- **Linear Polarization:** The electric field vector oscillates along a straight line. This occurs when the two orthogonal field components E_x and E_y are in-phase (phase difference $\Delta\psi = 0$ or 180°).
- **Circular Polarization:** The electric field vector traces out a circle. This requires $E_x = E_y$ and $\Delta\psi = \pm 90^\circ$ ($\pm\pi/2$ rad).
- **Elliptical Polarization:** The electric field vector traces an ellipse. This is the general case when $E_x \neq E_y$ and/or the phase difference is arbitrary (excluding linear/circular conditions).

Traces of Polarization States



5. Input Impedance, Efficiency, and Friis Transmission Equation

5.1 Input Impedance

The **input impedance** (Z_{in}) of an antenna is the impedance presented by the antenna at its terminals. It consists of a real part (input resistance R_{in}) and an imaginary part (input reactance X_{in}):

$$Z_{in} = R_{in} + jX_{in}$$

The input resistance R_{in} is composed of:

$$R_{in} = R_r + R_L$$

where R_r is the **radiation resistance** (representing the power converted into radiated waves) and R_L is the **loss resistance** (representing conduction and dielectric heating losses).

5.2 Antenna Efficiency

The total antenna efficiency (η_0) accounts for all losses in the system:

$$\eta_0 = \eta_r \eta_c \eta_d = \eta_r \eta_{cd}$$

where:

- $\eta_r = 1 - |\Gamma|^2$ is the reflection (mismatch) efficiency (Γ is the reflection coefficient).
- η_c is the conduction efficiency.
- η_d is the dielectric efficiency.
- $\eta_{cd} = \eta_c \eta_d = \frac{R_r}{R_r + R_L}$ is the radiation efficiency.

5.3 Friis Transmission Equation

The Friis transmission equation relates the power received by an antenna (P_r) to the power transmitted by another antenna (P_t), separated by a distance r in free space.

Friis Transmission Formula

Let G_t be the gain of the transmitting antenna and G_r be the gain of the receiving antenna. The received power is:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi r} \right)^2 (PLF)$$

where $PLF = |\hat{\rho}_t \cdot \hat{\rho}_r|^2$ is the Polarization Loss Factor ($0 \leq PLF \leq 1$). For matched polarizations, $PLF = 1$.

Derivation:

1. The power density W_t radiated by an isotropic source at distance r is:

$$W_{isotropic} = \frac{P_t}{4\pi r^2}$$

2. For a directive transmitting antenna with gain G_t , the power density in its direction of maximum radiation is:

$$W_t = \frac{P_t G_t}{4\pi r^2}$$

3. The receiving antenna collects this energy using its effective aperture A_{er} :

$$P_r = W_t A_{er} = \left(\frac{P_t G_t}{4\pi r^2} \right) A_{er}$$

4. Using the relationship between effective aperture and gain ($A_{er} = G_r \frac{\lambda^2}{4\pi}$):

$$P_r = \left(\frac{P_t G_t}{4\pi r^2} \right) \left(G_r \frac{\lambda^2}{4\pi} \right) = P_t G_t G_r \left(\frac{\lambda}{4\pi r} \right)^2$$

6. Radiation Integrals and Potential Functions

To calculate the fields radiated by an antenna, we use auxiliary potential functions rather than solving Maxwell's equations directly. The most common vector potentials are:

- $\vec{A}$ (Magnetic Vector Potential): Created by electric current sources $\vec{J}$.
- $\vec{F}$ (Electric Vector Potential): Created by magnetic current sources $\vec{M}$ (equivalent currents).

6.1 Relationship between Potentials and Fields

For a magnetic vector potential $\vec{A}$, the magnetic flux density is:

$$\vec{B} = \nabla \times \vec{A} \implies \vec{H} = \frac{1}{\mu} \nabla \times \vec{A}$$

From Maxwell's equation $\nabla \times \vec{E} = -j\omega\mu\vec{H}$, we substitute $\vec{H}$ to get:

$$\vec{E} = -j\omega\vec{A} - \frac{j}{\omega\mu\epsilon} \nabla(\nabla \cdot \vec{A})$$

6.2 Inhomogeneous Helmholtz Wave Equation

From Maxwell's equations, the potential function $\vec{A}$ satisfies:

$$\nabla^2 \vec{A} + k^2 \vec{A} = -\mu \vec{J}$$

where $k = \omega\sqrt{\mu\epsilon} = 2\pi/\lambda$ is the wave number.

The solution to this inhomogeneous wave equation in free space is:

$$\vec{A}(x, y, z) = \frac{\mu}{4\pi} \iiint_V \vec{J}(x', y', z') \frac{e^{-jkR}}{R} dV'$$

where:

- (x', y', z') are the coordinates of the source currents.
- (x, y, z) are the coordinates of the observation point.
- $R = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$ is the distance from the source point to the observation point.

In the far field, the distance R can be approximated as $R \approx r - r' \cos \psi$ for phase terms, and $R \approx r$ for amplitude terms. This simplifies the vector potential to:

$$\vec{A} \approx \frac{\mu e^{-jkr}}{4\pi r} \iiint_V \vec{J}(x', y', z') e^{jkr' \cos \psi} dV'$$

Quick Revision — Module 1

Fundamental Concepts of Antennas | PE-EC801A

Key Formulas and Metrics at a Glance

Radiation Intensity (U)	$U = r^2 W_{rad}$ (Watts/steradian)
Directivity (D)	$D = \frac{4\pi U}{P_{rad}}$
Gain (G)	$G = \eta_{cd} D = \frac{4\pi U}{P_{in}}$
Effective Aperture-Gain Relation	$A_{em} = \frac{\lambda^2}{4\pi} D_0$
Friis Transmission Equation	$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi r}\right)^2 (PLF)$
Helmholtz Equation Solution	$\vec{A} = \frac{\mu}{4\pi} \iiint \frac{\vec{J} e^{-jkR}}{R} dV'$
Helmholtz Wave Number (k)	$k = \frac{2\pi}{\lambda} = \omega \sqrt{\mu\epsilon}$

One-Line Definitions

- **Transducer:** A device that converts energy from one form to another; antennas convert guided electromagnetic waves into free-space waves.
- **Radiation Pattern:** The spatial distribution of physical quantities (like power or field intensity) radiated by an antenna.
- **HPBW:** The angle between the two directions in which the radiation intensity is half of the maximum.
- **Reactive Near-Field:** The region closest to the antenna where reactive non-radiating energy is stored.
- **Far-Field (Fraunhofer) Region:** The region where waves behave as transverse electromagnetic (TEM) plane waves and the pattern does not change with distance.
- **Reciprocity:** The property where transmitting and receiving characteristics of an antenna are identical.
- **Polarization Loss Factor (PLF):** A factor measuring mismatch loss between the polarization of a wave and the receiving antenna.

Frequently Asked / Viva Questions

Q1. What is the physical mechanism of antenna radiation?

Radiation is produced by accelerated or decelerated charges, or equivalently, by time-varying currents. Charges oscillating on a conductor generate changing electric and magnetic fields that propagate outward as electromagnetic waves.

Frequently Asked / Viva Questions (contd.)

Q2. State the three field regions of an antenna and write their boundary equations.

- Reactive near-field: $r < 0.62\sqrt{D^3/\lambda}$
- Radiating near-field (Fresnel): $0.62\sqrt{D^3/\lambda} \leq r < 2D^2/\lambda$
- Far-field (Fraunhofer): $r \geq 2D^2/\lambda$

Q3. Define directivity. How is it different from power gain?

Directivity is the ratio of radiation intensity in a given direction to average radiation intensity (purely directional focus). Power gain also accounts for efficiency (conduction and dielectric losses) in addition to directional focus ($G = \eta_{cd}D$).

Q4. State the reciprocity theorem for antennas.

The transmitting and receiving properties of an antenna (pattern, impedance, directivity) are identical. A voltage source V at the terminals of antenna A produces current I at antenna B , and the same source V at B produces the same current I at A .

Q5. What is Polarization Loss Factor (PLF)? What is its value for a vertically polarized antenna receiving a horizontally polarized wave?

PLF measures the loss due to polarization mismatch between the incoming wave and the receiving antenna. For orthogonal polarizations (vertical vs. horizontal), $PLF = 0$ (zero power is received).

Q6. Derive the relationship between effective aperture and directivity.

Under thermodynamic equilibrium, the ratio of maximum effective aperture to directivity is constant for all antennas: $\frac{A_{em}}{D_0} = \frac{\lambda^2}{4\pi}$. Thus, $A_{em} = \frac{\lambda^2}{4\pi} D_0$.

Q7. Explain the physical meaning of radiation resistance.

It is an equivalent resistance that would dissipate the same amount of power as heat as the antenna actually radiates into free space: $R_r = \frac{2P_{rad}}{I_0^2}$.

Q8. State the Friis transmission equation. What parameters does it relate?

$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi r}\right)^2$. It relates received power (P_r) to transmitted power (P_t) based on antenna gains (G_t, G_r), operating wavelength (λ), and separation distance (r).

Q9. What are auxiliary potential functions? Why are they used?

Auxiliary vector potentials ($\vec{A}$ and $\vec{F}$) are mathematical helper functions. They are used because it is much simpler to compute the potentials from the current distributions ($\vec{J}, \vec{M}$) and then take derivatives to get the fields ($\vec{E}, \vec{H}$), rather than solving Maxwell's equations directly.

Q10. Under what conditions does a charge radiate?

A charge radiates only if it is accelerated or decelerated. A stationary charge or a charge moving with constant velocity along a straight, infinite transmission line does not radiate.

Comparison of Field Regions

Property	Reactive Near-Field	Radiating Near-Field	Far-Field
Dominant Field Type	Reactive fields (energy storage)	Radiating fields (reactive fields decay)	Radiating fields (pure real power)
Pattern Shape	Changes drastically with distance	Varies with distance (r)	Independent of distance (r)
Phase Fronts	Highly complex	Non-spherical	Spherical (locally planar)
Wave Impedance	Complex (reactive)	Varies	Real ($\approx 377 \Omega$ in free space)